Equal Groups Explorer

TEACHER KEY

3rd Grade Math · Interactive Math · Explore & Reflect

Get started

Make equal groups or arrays to multiply, then share or group counters to divide. See multiplication, repeated addition and division equations side by side. Open the tool and use it to answer each question.

Explore

1. Show 5 groups of 3. Write a multiplication equation.
   5 × 3 = 15
2. Write 7 × 4 as repeated addition and find the product.
   4 + 4 + 4 + 4 + 4 + 4 + 4 = 28
3. How many counters are in an array with 3 rows of 8?
   24 (3 × 8 = 24)
4. Share 18 equally into 3 groups. How many are in each group?
   6 (18 ÷ 3 = 6)

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Explore more

5. Put 18 into groups of 3. How many groups?

6 groups (18 ÷ 3 = 6)

6. Find the unknown number: 4 × ? = 32

8, because 4 groups of 8 is 32 (32 ÷ 4 = 8)

Reflect

7. Reflect: How are sharing and grouping different? How are they the same?

Sharing knows the number of groups and finds the size; grouping knows the size and finds the number of groups. Both are division.

Teacher notes

- Ask students to tell a story for an equation before building it: “3 bags with 7 apples each.”
- Use the two division buttons with the same numbers to show that 24 ÷ 4 has two meanings and one answer.
- Totals that leave counters over set up the idea that some numbers cannot be shared equally.

Standards: 3.OA.A.1, 3.OA.A.2, 3.OA.A.3, 3.OA.B.6. Accept any correct answer where several are possible, and ask students to explain with the tool’s picture.